

DineFlash Buffet System – Functional Documentation

1. Introduction

This document defines the functional design and workflow of the **DineFlash Buffet System**, specifically tailored for buffet-style restaurants.

Key Focus Areas

- QR-based ordering
 - Utility-based food selection
 - Role-based access control
 - Buffet-specific ordering flow
 - Kitchen/order processing interface
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2. System Overview

The DineFlash Buffet System operates in **Buffet Mode**, which differs from traditional restaurant systems.

Buffet Mode Characteristics

- QR-based **Food on Demand**
- No traditional menu system
- Ordering based on **utilities (food counters)**

Examples of Utilities

- Omelette Counter
- Appam Counter
- Pasta Counter
- Noodles Counter

3. Platform Context & Product Flavours

The DineFlash Buffet System is implemented as a **separate product flavour** within the existing Flash platform.

3.1 Flash Platform Overview

The Flash platform is designed to support **multiple product flavours**, each tailored to different industry use cases while sharing a common core architecture.

This approach enables:

- Reuse of core modules (authentication, ordering engine, notifications)
- Customization of workflows and UI per use case
- Scalable and maintainable system design

3.2 Product Flavour Architecture

The existing Flash platform supports multiple **product flavours**, each customized for a different industry use case while sharing the same core architecture.

Current flavours include:

Food Flash

The standard restaurant and food service version used for managing food orders, queues, and service requests.

Airline Flash

A version designed for airport and airline environments to manage passenger service requests.

Dine Flash

A version focused on in-restaurant dining and table management.

DineFlash Buffet System (New Flavour Extension)

The **DineFlash Buffet System** is introduced as a **specialized extension of the Dine Flash flavour**, designed specifically for buffet-style restaurants.

3.3 Architectural Impact

- Reuses existing:

- User management
- Order lifecycle
- Notification system
- Introduces:
 - Utility-based ordering model
 - Combined customization flow
 - Utility User role
 - Kitchen parallel processing UI

3.4 Design Principle

The buffet flavour extends the Dine Flash system by modifying the **interaction model**, while preserving the underlying architecture.

4. QR Code System

4.1 Types of QR Codes

Dynamic QR Code

- Generated per session
- Can be reset
- More secure
- Recommended for real-time usage
- Dynamic QR codes help prevent reuse and enable session-based ordering by resetting access periodically.

Static QR Code

- Fixed QR placed on tables
- Static QR codes may be reused by customers (e.g., via photo capture), which can lead to unauthorized or delayed usage. This is accepted as a controlled trade-off if required.

4.2 QR Code Data Structure

Each QR code contains:

- **Outlet ID**
- **Table Number**

4.3 Scan Flow

1. Customer scans QR code
 2. System opens page using **Outlet ID**
 3. **Table number is auto-filled**
 4. Customer enters:
 - Name
 - Optional details (e.g., room number)
 5. Phone number can be disabled
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5. User Roles & Access

5.1 User Types

Role	Description
Admin Manager	Full access to system
Utility Manager	Access to assigned utilities
Utility User	Handles kitchen/order execution
General User	Customer

5.2 Permissions

Admin Manager

- Access all utilities
- View/send all chats

Utility Manager

- Access assigned utilities
- Manage specific operations
- View/send chats of assigned users

Utility User

- View incoming orders
- Process orders
- Mark items as ready

5.3 Order Visibility Rule

- Orders are not intended for Admin Manager or Utility Manager views for execution purposes.
 - Only Utility Users handle actionable orders.
 - Admin Manager and Utility Manager may have visibility for monitoring purposes, but they are not part of the execution workflow.
 - All real-time order processing and status updates are handled exclusively by Utility Users.
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6. Utility-Based Ordering System

6.1 Concept

Traditional menus are replaced with:

Utilities = Food Counters

Each utility represents a physical food counter (e.g., omelette station, appam counter), and users are mapped accordingly to handle orders from those specific counters.

6.2 Structure

Utility
└─ Options (Customization)

6.3 Ordering Flow

1. Customer selects utilities
 2. Customer sets quantity (+ / -)
 3. No customization at this stage
 4. Customer proceeds to submission
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7. Cart & Submission Flow

Approach: Final Submission Options

The system follows a **single standardized flow**.

7.1 Flow Description

1. User selects items (utilities)
 2. User sets quantity for each item
 3. User clicks **Proceed / Submit**
 4. System shows **combined options screen**
 5. User customizes each item
 6. Final order is submitted
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7.2 Options Handling

- All items appear in one screen
- Each quantity expands into individual entries

Example

If user selects:

- Omelette × 2
- Appam × 3

System shows:

- Omelette 1 → options
 - Omelette 2 → options
 - Appam 1 → options
 - Appam 2 → options
 - Appam 3 → options
-

7.3 Options Types

Predefined Options (Primary)

- Checkbox-based:
 - No onion
 - No capsicum
 - No pepper

Remarks (Optional)

- Free text:
 - “Less oil”

- “Extra spicy”
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7.4 Key Characteristics

- No popups during item selection
 - Single structured customization step
 - Faster and cleaner user experience
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7.5 Advantages

- Reduces ordering friction
 - Improves usability in buffet environments
 - Ensures clarity for kitchen staff
 - Avoids repeated interruptions
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8. Order Processing Flow

8.1 Problem in Traditional Flow

Orders going to:

- Admin Manager
- Utility Manager

Not useful for kitchen execution

8.2 Solution: Utility User Role

- Orders are sent directly to **Utility Users**
 - Acts as kitchen/counter interface
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9. Utility User Interface (Kitchen View)

9.1 Order Display

Each order appears as a card:

- Order Number
- Table Number
- Number of items

9.2 Order Details

On selecting an order:

- List of items
 - Quantity
 - Customization options
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9.3 Order Completion

- Utility user clicks **“Ready”**
 - Customer receives notification
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10. Multi-Utility Mapping

Each utility corresponds to a physical counter in the restaurant, and mapping ensures that orders are routed to the correct preparation station.

Problem

Multiple counters exist:

- Appam
 - Omelette
 - Pasta
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Solution

- Each utility user is mapped to specific utilities
 - One user can handle:
 - Single utility OR
 - Multiple utilities
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11. Kitchen Display Optimization

11.1 Concurrent Orders

- The system supports parallel processing of multiple orders, reflecting real kitchen operations where multiple dishes are prepared simultaneously.
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11.2 Layout Options

Grid View

- Display 4–5 orders simultaneously
 - New orders replace completed ones
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11.3 Screen Layout

- **Top 70–80%** → Order details
 - **Bottom 20–30%** → Queue/navigation
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12. Reservation Handling

- Current:
 - Supports **current reservations**
 - Future:
 - Separate **table reservation module**
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13. Key Configuration Options

- Enable/Disable **Single Utility View**
 - Auto-select utility (if only one exists)
 - Static vs Dynamic QR configuration
 - Enable/Disable utility options
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14. Summary of Key Changes

Major Additions

- Utility Options (Customization system)

- Utility Mapping
 - Utility User Role (Kitchen role)
 - QR with Table + Outlet ID
 - Final Submission Customization Flow
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15. Conclusion

The DineFlash Buffet System transforms ordering from:

Menu-based ordering to
Utility-based real-time ordering

Core Benefits

- Faster customer interaction
 - Simplified UI/UX
 - Efficient kitchen workflow
 - Flexible and scalable design
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DineFlash Buffet System – UI Wireframes

1. QR Scan → User Details Screen

Purpose

Capture minimal user info + auto-fill table

Wireframe

Restaurant Name	
Table No: 12 (Auto-detected from QR)	
Name:	
Room No:	
Start Ordering	

UX Notes

- Keep it minimal (buffet = fast)
 - No login/signup
 - Phone number optional or hidden
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2. Utility Selection Screen (Main Screen)

Purpose

Customer selects food items (utilities)

Wireframe

	Select Your Items			

	Omelette	[-]	0	[+]
	Appam	[-]	0	[+]
	Pasta	[-]	0	[+]
	Noodles	[-]	0	[+]

	[Proceed]			

UX Notes

- Only quantity selection
 - Fast interaction
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3. Combined Options Screen (CORE SCREEN)

Purpose

All customization happens here (after clicking Proceed)

	Customize Your Order	

	Table: 12 Name: Archa	

Omelette (2)	

Omelette 1	
[] No Onion	
[] No Capsicum	
[] No Tomato	
Remark: [_____]	

Omelette 2	
[] No Onion	
[] Extra Cheese	
Remark: [_____]	

Appam × 3	
(No customization required)	

[Submit Order]	

UX Notes (VERY IMPORTANT)

- Single scrollable screen
 - Each item = separate block
 - No repeated popups
 - This is the **core differentiator of your system**
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4. Order Confirmation Screen

Purpose

Show success + order ID

Wireframe

Order Placed Successfully	
Order No: #102	
Table: 12	
Your food is being prepared	

5. Multi-Order Kitchen View (Parallel Cooking)

Purpose

Handle multiple orders simultaneously

ACTIVE WORK AREA				
Order #102	Order #103	Order #104	Order #105	
Table 12	Table 8	Table 5	Table 2	
Omelette 2	Pasta 1	Appam 3	Noodles 1	
- No onion	- Cheese		- No spice	
- No caps.				
[READY]	[READY]	[READY]	[READY]	
ORDER QUEUE				

	#106		T3		2 items		#107		T9		1 item		#108		T4	
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UX Notes

- Designed for speed
- Each column = one active order
- New orders replace completed ones

8. Customer Notification (After Ready)

Your order (#102) is ready for pickup

9. Admin – Utility Mapping Screen

Purpose

Assign counters to kitchen staff

Wireframe

	Create Utility User	
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	Name: [_____]	
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	Assign Utilities:	
	[☒] Omelette Counter	
	[☒] Appam Counter	
	[☐] Pasta Counter	
	[☐] Noodles Counter	

	[Save]	
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End-to-End Flow

QR Scan



User Details



